

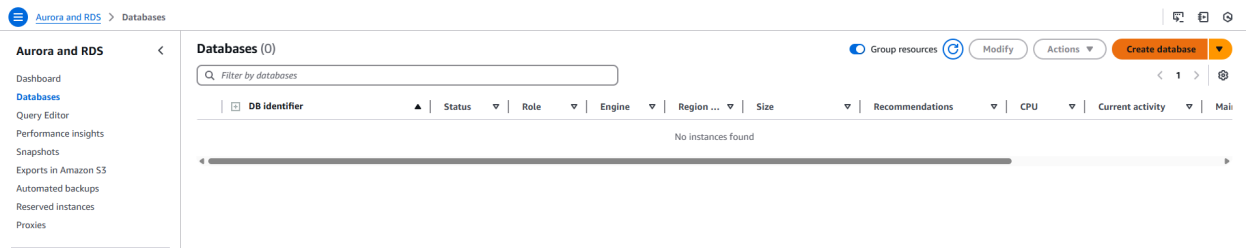
Prerequisites:

- AWS CLI installed & configured (aws configure)
- Docker installed (for building images)
- Snipe-IT environment variables ready (DB credentials, APP key, etc.)
- An RDS (MySQL/MariaDB) database for persistent storage

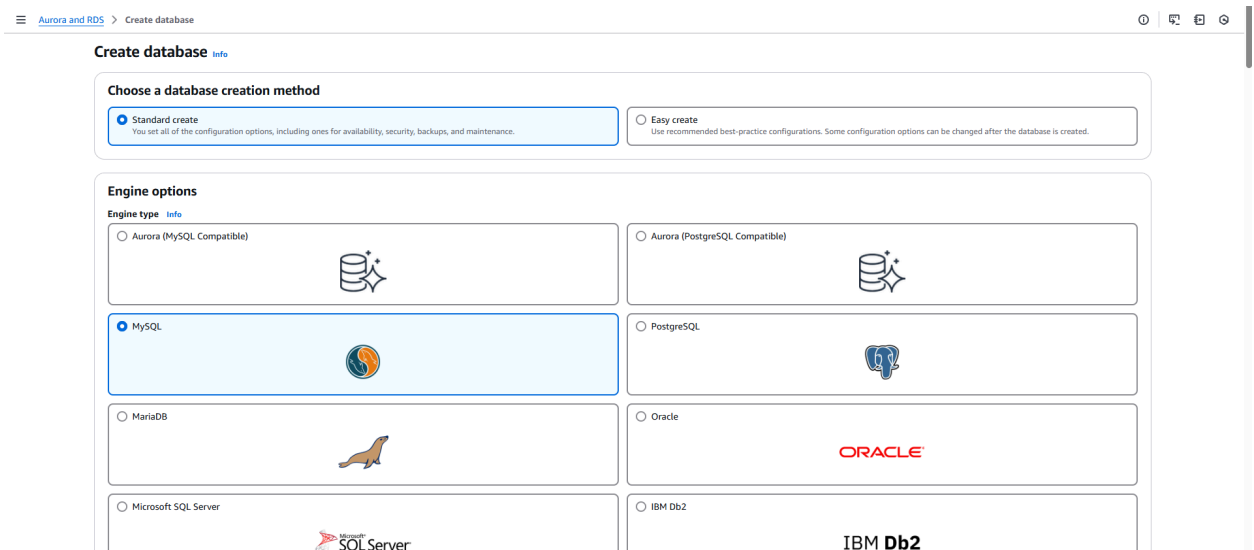
Create an RDS Database (MySQL/MariaDB)

Fargate is stateless, we need a persistent database. You can create an AWS RDS instance:

- Go to AWS RDS Console → Click Create database



- Choose MySQL or MariaDB



- Set instance type as per your application usage (e.g., `db.t3.micro` for small deployments)
- Set database name, username, and password

Settings

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

snipe-it-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ **Credentials Settings**

Master username [Info](#)
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ **Managed in AWS Secrets Manager - most secure**
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ **Auto generate password**
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Weak](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

- Enable Public access (or configure security groups properly)
- Click Create after all configuration.
- Take note of the RDS Endpoint for later.

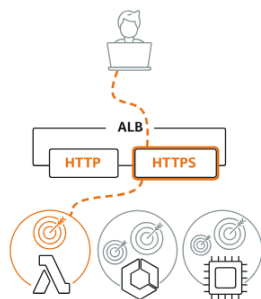
Set Up an Application Load Balancer (ALB)

- Go to EC2 Console → Load Balancers
- Click Create Load Balancer → Application Load Balancer

EC2 > Load balancers > Compare and select load balancer type

Load balancer types

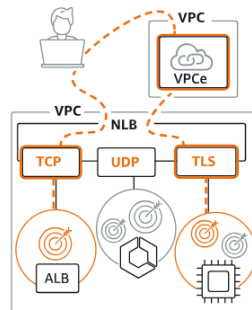
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

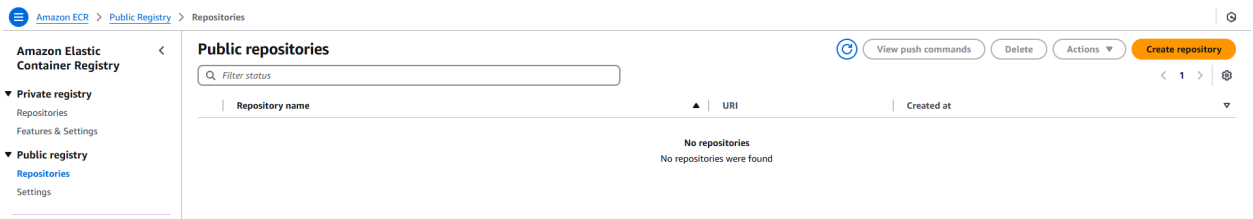
[Create](#)

► **Classic Load Balancer - previous generation**

- Name: `snipe-it-alb`
- Listeners: Port `80`
- Create Target Group: Go to the **AWS Console** → **EC2** → **Load Balancers** → **Target Groups** and create a new target group
 - Target type: IP (since Fargate uses dynamic IPs)
 - Protocol: HTTP or HTTPS (depending on SSL setup)
 - Port: 80 (or 443 if using HTTPS)
 - VPC: Select the VPC where your ECS service will run
 - Health check protocol: HTTP
 - Health check path: `/health` (or `/` if Snipe-IT doesn't have a health endpoint)
 - Healthy threshold: 3
 - Unhealthy threshold: 2
 - Timeout: 5 seconds
 - Interval: 30 seconds
- Click Create

Create an ECR Repository (AWS Elastic Container Registry)

- Go to AWS ECR Console → Click Create Repository



- Set repository name

Create public repository

Details

Repository name Info
 A namespace can be included with your repository name (e.g. namespace/repo-name).
 public.ecr.aws/t0r1w8c7/
8 out of 205 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./.

Repository logo - optional Info
 Choose a local image file in PNG format to use as the repository logo.

The supported file format is PNG. The supported image dimensions for both height and width should be a minimum of 60 pixels and a maximum of 2048 pixels. The maximum file size is 500 KB.

Short description - optional Info
 The short description is displayed in search results and on the repository detail page.

33 out of 255 characters maximum.

Content type - optional Info
 Select the operating systems and system architectures that are compatible with the images in your repository.

Architectures
☐ ARM
☐ ARM 64
☐ x86
☐ x86-64

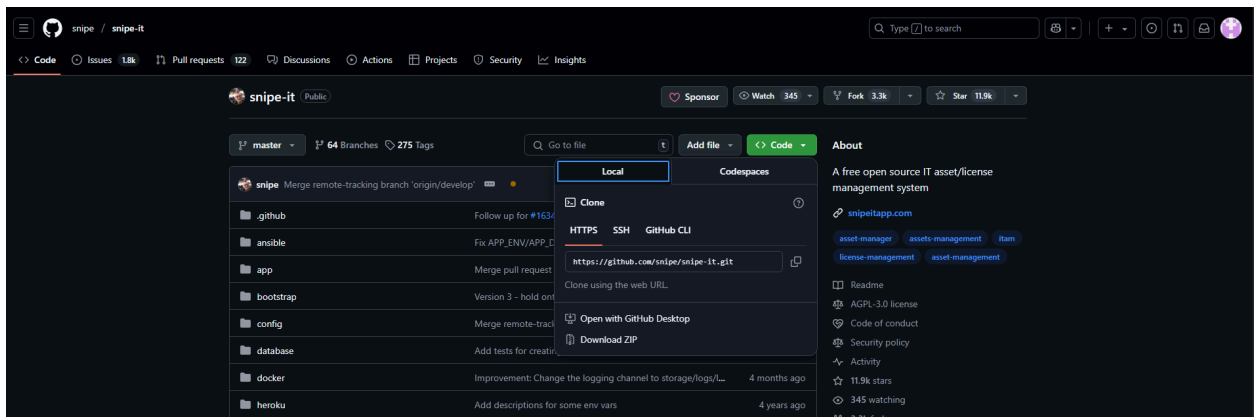
Operating systems
☐ Linux
☐ Windows

- Click Create

Build and Push the Snipe-IT Docker Image

Clone Snipe-IT

- Go to the browser and search Snipe-IT Github
- Click first github hub link
 - <https://github.com/snipe/snipe-it>
- Click code and copy the link
 - <https://github.com/snipe/snipe-it.git>



- Go to the CLI and clone this repository by using this following command `git clone https://github.com/snipe/snipe-it.git`
- List out all the directories using `ls` command check our repository snipe-it cloned successfully.

```
ankul@Superuser:~$ git clone https://github.com/snipe/snipe-it.git
Cloning into 'snipe-it'...
remote: Enumerating objects: 220355, done.
remote: Counting objects: 100% (25/25), done.
remote: Compressing objects: 100% (24/24), done.
remote: Total 220355 (delta 7), reused 4 (delta 1), pack-reused 220330 (from 2)
Receiving objects: 100% (220355/220355), 222.19 MiB | 1.66 MiB/s, done.
Resolving deltas: 100% (148200/148200), done.
Updating files: 100% (7545/7545), done.
ankul@Superuser:~$ ls
aws  awscli2.zip  docker-compose.yml  ecomm  get-docker.sh  snipe-it
ankul@Superuser:~$ |
```

- In the Snipe-IT repository, create a `.env` file to configure application settings. This file will hold the necessary environment variables like database connection, mail server settings, and others.
- Example `.env` file:

```
APP_ENV=production
APP_DEBUG=false
APP_URL=http://your-load-balancer-dns
APP_KEY=base64:your-app-key-here
```

```
DB_CONNECTION=mysql
DB_HOST=your-rds-endpoint
DB_PORT=3306
DB_DATABASE=your-db-name
DB_USERNAME=your-db-username
DB_PASSWORD=your-db-password
```

```
MAIL_MAILER=smtp
MAIL_HOST=smtp.mailtrap.io
MAIL_PORT=587
MAIL_USERNAME=your-email-username
MAIL_PASSWORD=your-email-password
MAIL_ENCRYPTION=tls
```

SESSION_DRIVER=database

CACHE_DRIVER=redis

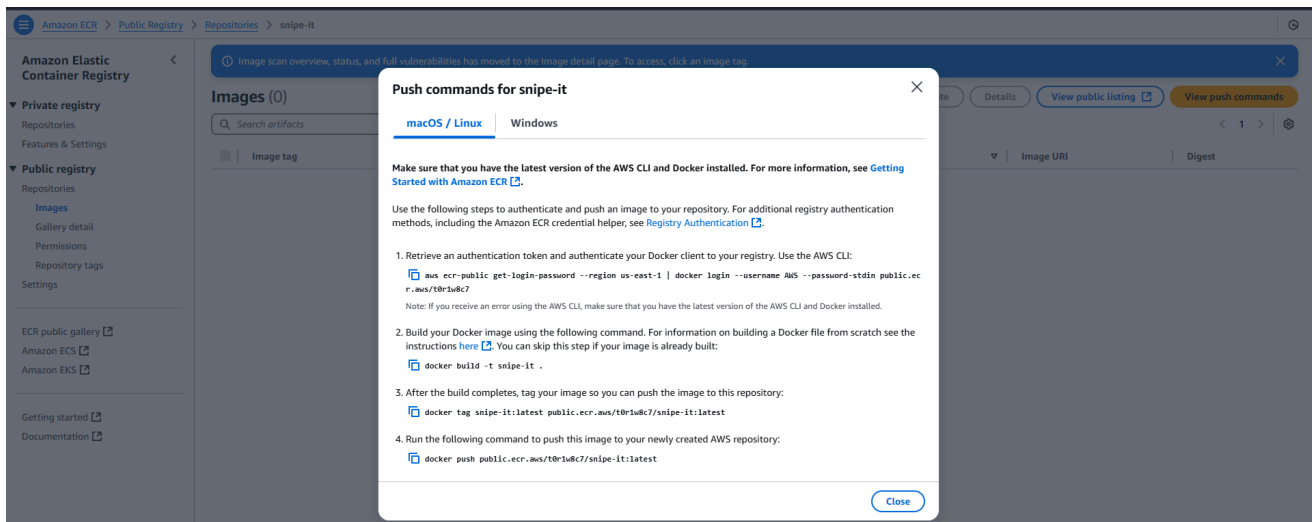
REDIS_HOST=your-redis-endpoint

REDIS_PASSWORD=null

REDIS_PORT=6379

Docker with ECR (Authenticate, Build Docker Image, Tag and Push Image)

- Go to the ECR repository snipe-it and click view push commands in upper right hand side



Here you find all the commands which will help you to authenticate and build your docker image and please make sure you also read this document carefully once.

```
ankul@Superuser:~/snipe-it$ aws ecr-public get-login-password --region us-east-1 | docker login --username AWS --password-stdin public.ecr.aws/t0rlw8c7
WARNING! Your credentials are stored unencrypted in '/home/ankul/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ankul@Superuser:~/snipe-it$ docker build -t snipe-it .
[+] Building 100.2s (35/35) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default
=> => transferring dockerfile: 4.28kB                                           0.0s
=> [internal] load metadata for docker.io/library/composer:latest                0.0s
=> [internal] load metadata for docker.io/library/ubuntu:24.04                  2.0s
=> [internal] load .dockerignore                                                  2.2s
=> [internal] load .dockerignore                                                  0.0s
```

```

=> => unpacking to docker.io/library/snipe-it:latest
ankul@Superuser:~/snipe-it$ docker tag snipe-it:latest public.ecr.aws/t0rlw8c7/snipe-it:latest
ankul@Superuser:~/snipe-it$ docker push public.ecr.aws/t0rlw8c7/snipe-it:latest
The push refers to repository [public.ecr.aws/t0rlw8c7/snipe-it]
7f402d0cc9c4: Pushed
1b601d1babbf: Pushed
f92f5d778cfd: Pushed
d0fc23e3436d: Pushed
0f5a9b2db198: Pushed
8742c7ef2ac7: Pushed
4f4fb700ef54: Pushed
6eed15b551a2: Pushed
5a7813e071bf: Pushed
7dc95af578b0: Pushed
d2f149326299: Pushed
ac779ec2a086: Pushed
c07d8dcddbca: Pushed
7528939d7f7d: Pushed
1ae282087fb4: Pushed
a06ad1d1f294: Pushed
2adec9786aee: Pushed
5aa8be8a208e: Pushed
72ab380d3a68: Pushed
9c0e211a3702: Pushed
2722b5eee9d6: Pushed
b251c6a8f785: Pushed
e913f3f5c5e1: Pushed
a27406d58434: Pushed
436b04db17dc: Pushed
e43ba748b920: Pushed
4ad7d55d2089: Pushed
5e324fe0f2b2: Pushed
fd31e0e0a18d: Pushed
latest: digest: sha256:bcc55b572560ff850263ec371550c322009ce88b395599b0f19dad325d796f2c size: 856

```

After command execution check your repository your docker image push successfully or not.

The screenshot shows the Amazon ECR console for the 'snipe-it' repository. The left sidebar shows the navigation menu with 'Public registry' selected. The main area displays 'Images (3)' with a table of artifacts.

Image tag	Artifact type	Pushed at	Size (MB)	Image URI	Digest
latest	Image Index	01 April 2025, 18:12:27 (UTC+05:5)	338.82	Copy URI	sha256:bcc55b572560ff850263ec371550c3...
-	Image	01 April 2025, 18:12:26 (UTC+05:5)	0.00	Copy URI	sha256:ecda5e23cf0ac8e12b065684fd6a5a...
-	Image	01 April 2025, 18:12:26 (UTC+05:5)	338.82	Copy URI	sha256:a1e96b074fb5245d83006443e6248...

Note: If you receive an error using the AWS CLI, make sure that you have the latest version of the AWS CLI and Docker installed.

Create an ECS Cluster

- Go to ECS Console → Clusters → Click Create Cluster

The screenshot shows the Amazon ECS console 'Clusters' page. The left sidebar shows the navigation menu with 'Clusters' selected. The main area displays 'Clusters (0)' with a search bar and a table of clusters. The table is empty, showing 'No clusters' and 'No clusters to display'.

- Select AWS Fargate and name snipe-it-cluster

Amazon Elastic Container Service > Create cluster

Cluster configuration

Cluster name
snipe-it-cluster
Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

► **Service Connect defaults - optional**

▼ **Infrastructure - optional** [Info](#) Serverless
Your cluster is automatically configured for AWS Fargate (serverless) with two capacity providers. Add Amazon EC2 instances.
☒ **AWS Fargate (serverless)**
Pay as you go. Use if you have tiny, batch, or burst workloads or for zero maintenance overhead. The cluster has Fargate and Fargate Spot capacity providers by default.
☐ **Amazon EC2 instances**
Manual configurations. Use for large workloads with consistent resource demands.
 ⓘ External instances using **ECS Anywhere** can be registered after cluster creation is complete.

► **Monitoring - optional** [Info](#)
CloudWatch Container Insights is a monitoring and troubleshooting solution for containerized applications and microservices.

► **Encryption - optional** [Info](#)
Choose the KMS keys used by tasks running in this cluster to encrypt your storage.

► **Tags - optional** [Info](#)
Tags help you to identify and organize your clusters.

[Cancel](#) [Create](#)

- Click Create

Create a Task Definition

In your ECS Task Definition, you need to define environment variables for the container.

- Go to **ECS Console** → **Task Definitions** → **Create new task**

Amazon Elastic Container Service > Task definitions

Task definitions (0) [Info](#)

Last updated April 01, 2025 at 18:30 (UTC+5:30) [Deploy](#) [Create new revision](#) [Create new task definition](#)

Filter by status: Active

Search: Filter task definitions

Task definition | Status of last revision

No task definitions
No task definitions to display.
[Create new task definition](#)

- Launch type → Select Fargate

Infrastructure requirements
Specify the infrastructure requirements for the task definition.

Launch type [Info](#)
Selection of the launch type will change task definition parameters.
☒ **AWS Fargate**
Serverless compute for containers.
☐ **Amazon EC2 instances**
Self-managed infrastructure using Amazon EC2 instances.

OS, Architecture, Network mode
Network mode is used for tasks and is dependent on the compute type selected.

Operating system/Architecture [Info](#)
Linux/X86_64

Network mode [Info](#)
awsvpc

Task size [Info](#)
Specify the amount of CPU and memory to reserve for your task.

CPU
1 vCPU

Memory
3 GB

▼ **Task roles - conditional** [Info](#)
Task role [Info](#)
A task IAM role allows containers in the task to make API requests to AWS services. You can create a task IAM role from the [IAM console](#).
 -

Task execution role [Info](#)
A task execution IAM role is used by the container agent to make AWS API requests on your behalf. If you don't already have a task execution IAM role created, we can create one for you.
 ecstask

- Container:

Container - 1 [Info](#)

Container details
Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name **Image URI**

Essential container ☒ Yes [Remove](#)

Private registry [Info](#)
Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.
☒ Private registry authentication

Port mappings [Info](#)
Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port	Protocol	Port name	App protocol
80	TCP	container-port-protocol	HTTP

[Add port mapping](#) [Remove](#)

Read only root file system [Info](#)
When this parameter is turned on, the container is given read-only access to its root file system.
☐ Read only

Resource allocation limits - conditional [Info](#)
Container-level CPU, GPU, and memory limits are different from task-level values. They define how much resources are allocated for the container. If container attempts to exceed the memory specified in hard limit, the container is terminated.

CPU	GPU	Memory hard limit	Memory soft limit
1 in vCPU	1	3 in GB	7 in GB

Environment variables - optional
[Info](#)
Environment variables
Add individually
Add a key-value pair to specify an environment variable.

- Container Name: Your Container name
- Image URL: Paste here ECR repository url which we created early
- Environment Variables:** Set environment variables same as snipe-it.env file here.

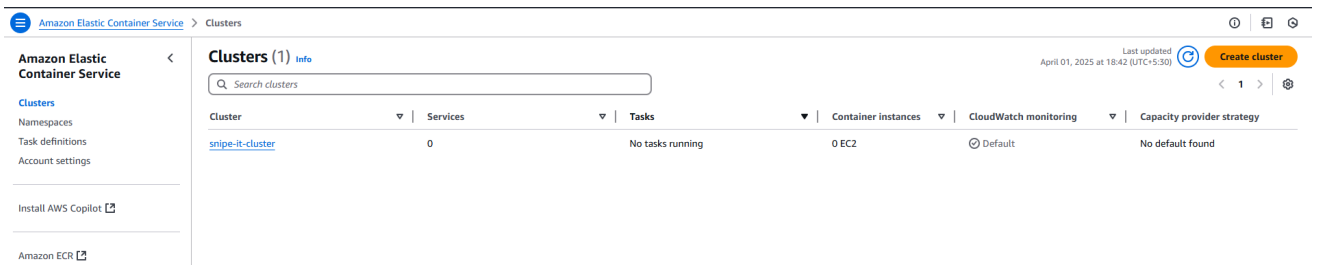
Environment variables - optional
[Info](#)
Environment variables
Add individually
Add a key-value pair to specify an environment variable.

Key	Value type	Value
APP_ENV	Value	production
APP_DEBUG	Value	false
APP_URL	Value	http://snipe-it-alb-81047319.us-
APP_KEY	Value	base64:0qXFc3PLtFC3clGMnyO7J
DB_CONNECTION	Value	mysql
DB_HOST	Value	snipe-it-db.c3g4mi64yq52.us-we
DB_PORT	Value	3306
DB_DATABASE	Value	snipe-it-db
DB_USERNAME	Value	admin
DB_PASSWORD	Value	Ankul@123
MAIL_DRIVER	Value	smtp
MAIL_HOST	Value	smtp.gmail.com

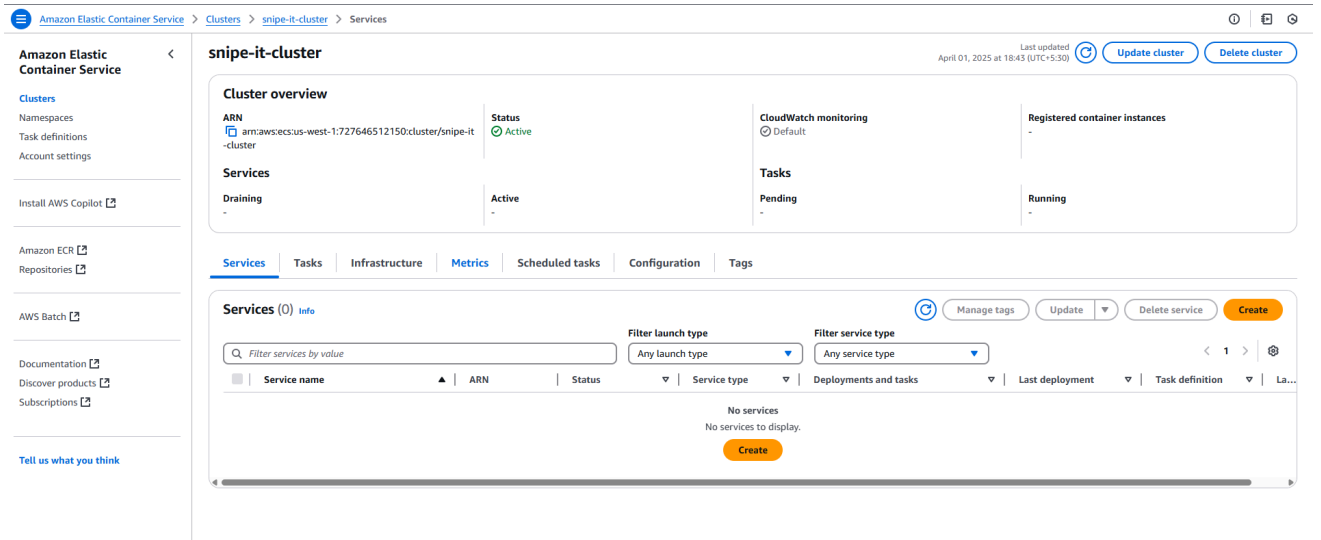
- Click **Create**

Create a Service in ECS

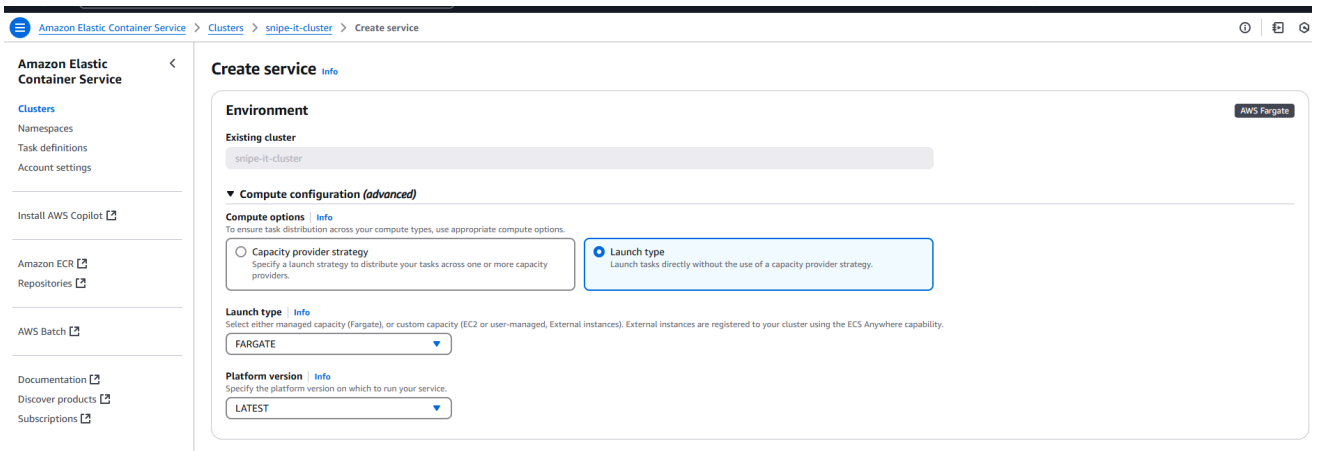
- Go to ECS Console → Clusters → Select Cluster snipe-it-cluster



- Click Create Service



- Launch Type → Select Fargate



- Task Definition → Choose the task you created

Amazon Elastic Container Service > Clusters > snipe-it-cluster > Create service

Amazon Elastic Container Service

- Clusters
- Namespaces
- Task definitions
- Account settings

Install AWS Copilot

Amazon ECR
Repositories

AWS Batch

Documentation
Discover products
Subscriptions

Tell us what you think

Deployment configuration

Task definition family
Select an existing task definition family. To create a new task definition, go to [Task definitions](#).

Snipe-IT

Task definition revision
Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.

5

Service name
Assign a service name that is unique for this cluster.
Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster.

Snipe-IT-Service

Service type | Info
Specify the service type that the service scheduler will follow.

☒ Replica
Place and maintain a desired number of tasks across your cluster.

☐ Daemon
Place and maintain one copy of your task on each container instance.

Desired tasks
Specify the number of tasks to launch.

2

Availability Zone rebalancing | Info
☒ Turn on Availability Zone rebalancing
Amazon ECS automatically detects Availability Zone imbalances in task distributions across an ECS service, and evenly redistributes ECS service tasks across Availability Zones.

Health check grace period | Info
5
seconds

► Deployment options

- Desired Tasks → Set to as per usage (for now 2)
- Networking: Select a VPC & Subnet (must allow outbound internet)

Amazon Elastic Container Service > Clusters > snipe-it-cluster > Create service

Amazon Elastic Container Service

- Clusters
- Namespaces
- Task definitions
- Account settings

Install AWS Copilot

Amazon ECR
Repositories

AWS Batch

Documentation
Discover products
Subscriptions

Tell us what you think

Networking

VPC | Info
Select a VPC to use for your Amazon ECS resources.

vpc-09d4cbf7ce4c976da
default

Create a new VPC

Subnets
Choose the subnets within the VPC that the task scheduler should consider for placement.

Choose subnets

subnet-04cbfdae0438fcad
us-west-1c 172.31.16.0/20

subnet-02c3fba3d8082213
us-west-1b 172.31.0.0/20

Clear current selection

Security group | Info
Choose an existing security group or create a new security group.

☒ Use an existing security group
☐ Create a new security group

Security group name
Choose an existing security group.

Choose security groups

sg-0bac4b3b957463126
test

sg-09378bf6c33242468
default

Public IP | Info
Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).

☒ Turned on

- Load Balancing: Configure your load balancer which one you created early.

Amazon Elastic Container Service

Clusters

Namespaces

Task definitions

Account settings

Install AWS Copilot

Amazon ECR

Repositories

AWS Batch

Documentation

Discover products

Subscriptions

Tell us what you think

Clusters

snipe-it-cluster

Create service

▼ Load balancing - optional

Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.

☒ Use load balancing

VPC

The VPC for your load balancing resources must be the same as the VPC for your service with awswpc.

vpc-09d4cbf7ce4c976da

Load balancer type

Specify the load balancer type to distribute incoming traffic across the tasks running in your service.

☒ Application Load Balancer

An Application Load Balancer makes routing decisions at the application layer (HTTP/HTTPS), supports path-based routing, and can route requests to one or more ports.

☐ Network Load Balancer

A Network Load Balancer makes routing decisions at the transport layer (TCP/UDP).

Container

The container and port to load balance the incoming traffic to

Snipe-IT 80:80

Host port:Container port

Application Load Balancer

Specify whether to create a new load balancer or choose an existing one.

☐ Create a new load balancer

☒ Use an existing load balancer

Load balancer

Choose an existing load balancer to distribute traffic. View exiting load balancers and create new one in EC2 Console

snipe-it-alb

snipe-it-alb-81047319.us-west-1.elb.amazonaws.com

internet-facing

Listener

Specify the port and protocol that the load balancer will listen for connection requests on.

☐ Create new listener

☒ Use an existing listener

Listener

80:HTTP

- Click Create Service